Trading System Signal & Indicator Specifications

1. Overview

This document defines the quantitative signals and indicators calculated for Polymarket's 5-minute crypto Up/Down markets (for example btc-updown-5m-<t>): what each signal represents, how it is derived from 1-second raw snapshots, and why it is required by the strategy.

- **Purpose:** Raw order books and prices are too noisy for direct execution. Indicators convert high-frequency ticks into clean mathematical signals governing **when to enter**, **which side to pick (UP vs DOWN)**, and **when to exit**.
- **Streamlined Architecture:** We only compute indicators directly tied to execution rules: Price Gap, Wilder RSI(14) on gap, Wilder ATR(14) on BTC, Real-Time Volatility (RTV), Gap Safety Ratio, Peak Gap Ratio, and Order Book Imbalance (OBI).
- **Execution Horizon:** Markets resolve strictly after 300 seconds (5 minutes). Signals are updated once every second to detect entry windows within the final 90 seconds.

2. Indicator Inventory

Indicator	Input Data	Range / Unit	Role in Trading Strategy
`price_gap`	Binance BTC vs Oracle Open	Dollar ($)	Primary directional lead signal.
`rsi_gap`	14-tick gap delta	0 to 100	Gap momentum & stop-loss suppression.
`atr_btc`	14-tick BTC price delta	Dollar ($)	Volatility filter (ATR ≤ $2.0 required).
`rtv_btc`	30-tick rolling returns	Dollar ($)	Micro-volatility tick noise filter.
`gap_safety`	|gap| / ATR	Ratio	Directional cushion (Safety ≥ 40 required).
`peak_gap_ratio`	|gap| / peakAbsGap	0.0 to 1.0	Trend retention filter (Ratio ≥ 0.75 required).
`up_obi / down_obi`	Top-5 L2 depth bids/asks	-1.0 to +1.0	Order book liquidity pressure.
`up_midpoint`	Best bid & ask midpoint	0.00 to 1.00	Market implied probability of UP outcome.

3. Signal Details

3.1 Price Gap (price_gap)

```
{
  "chainlink_price_to_beat": 75468.25,
  "binance_current_price": 75585.40,
  "price_gap": 117.15,
  "price_gap_abs": 117.15,
  "price_gap_pct": 0.155,
  "gap_direction": "UP"
}
```

Field	Meaning
`chainlink_price_to_beat`	The Chainlink BTC/USD price at window open. The final price must finish above/below this price.
`binance_current_price`	Real-time BTC/USDT price from Binance ticker stream.
`price_gap`	Current dollar lead (Binance price minus price to beat). Positive = UP leading; Negative = DOWN leading.
`gap_direction`	Direction of the lead ('UP' if positive, 'DOWN' if negative).

Why we need it: The price gap is the primary directional signal. At resolution time, whichever side has the lead wins the $1.00 payout. Measuring this lead in real time tells the bot which side is favored.

3.2 Wilder RSI on Price Gap (rsi_gap)

```
{
  "period": 14,
  "rsi_gap": 62.45,
  "momentum": "EXPANDING"
}
```

Field	Meaning
`period`	14 ticks (14 seconds of price gap updates).
`rsi_gap`	Wilder Relative Strength Index of the gap (0 to 100). >50 means gap is expanding; <50 means contracting.

Why we need it: RSI confirms that a price lead is gaining momentum rather than slowing down. Crucially, it acts as a **stop-loss suppressor**: if an entered position experiences a minor pullback but RSI remains strong (>50 for UP), the bot suppresses panic selling, preventing costly false exits.

3.3 Wilder ATR on Asset Price (atr_btc)

```
{
  "period": 14,
  "atr_btc": 1.45,
  "regime": "LOW_VOLATILITY"
}
```

Field	Meaning
`atr_btc`	Average true range (dollar volatility per second) over the past 14 seconds.

`regime`	Volatility regime filter. Late entry requires calm markets with ATR ≤ $2.00.

Why we need it: A price gap is only safe when volatility is low. If BTC is swinging by $20/second, even a $50 lead can flip in three seconds. Requiring ATR ≤ $2.0 ensures we only trade in stable market conditions.

3.4 Gap Safety Ratio (gap_safety)

```
{
  "price_gap_abs": 92.50,
  "atr_btc": 1.45,
  "gap_safety": 63.79,
  "entry_eligible": true
}
```

Field	Meaning
`gap_safety`	The ratio |price_gap| / ATR. Measures how many volatility units the lead covers.
`entry_eligible`	true when gap_safety ≥ 40.0. Indicates strong statistical cushion.

Why we need it: Gap Safety is our primary trade filter. A gap of $80 with an ATR of $1.50 gives a safety ratio of 53.3, meaning the price lead is over 50x larger than normal 1-second price fluctuations. Requiring safety ≥ 40.0 guarantees an overwhelming directional cushion.

3.5 Peak Gap Ratio (peak_gap_ratio)

```
{
  "current_gap": 85.00,
  "peak_abs_gap": 95.00,
  "peak_gap_ratio": 0.895,
  "trend_retained": true
}
```

Field	Meaning
`peak_abs_gap`	The largest absolute lead observed during this 5-minute round.
`peak_gap_ratio`	Current gap divided by peak gap (0.0 to 1.0). Late entry requires ratio ≥ 0.75.

Why we need it: Prevents buying into collapsing leads. If a lead reached $100 earlier but has dropped to $25, the trend has decayed. Requiring ratio ≥ 0.75 guarantees that the lead is maintaining at least 75% of its peak power.

3.6 Order Book Imbalance & Midpoint (up_obi, up_midpoint)

```
{
  "up_best_bid": 0.88,
  "up_best_ask": 0.90,
  "up_midpoint": 0.89,
  "up_bid_liquidity": 12450.0,
  "up_ask_liquidity": 3200.0,
  "up_obi": 0.591
}
```

Field	Meaning
`up_obi`	Imbalance from -1.0 to +1.0. Positive = deep bid support; Negative = heavy ask resistance.
`up_midpoint`	Midpoint price (0.00 to 1.00). Reflects market's implied probability of winning.

Why we need it: Order book imbalance detects liquidity backing before price moves. A positive OBI confirms that market makers are bidding aggressively, providing execution liquidity for our orders.

4. Strategy Execution Rules

The calculated indicators govern the entry and exit decisions of our two strategy models:

Strategy Model	Execution Conditions	Risk Management & Exit
Late-Entry Strategy	• Remaining time ≤ 90 seconds • atr_btc ≤ 2.0 (calm regime) • gap_safety ≥ 40.0 (wide cushion) • peak_gap_ratio ≥ 0.75 (strong retention) • Target ask price > 0.85 • Available counterparty liquidity ≥ $20	• Initial stop-loss set at $0.48. • Stop-loss suppressed if rsi_gap and lead direction confirm continuation. • Emergency market FOK sell triggered on order expiry.
Limit Order Simulation	• Places maker limit buy at $0.49 immediately at window open. • Unfilled buy expires 100s before close.	• Upon buy fill, places take-profit sell at $0.70. • Take-profit expires at T-30s. • Emergency market sell exits position at T-30s if unfilled.

5. Conclusion & Next Step

By filtering raw 1-second ticks into these 6 quantitative indicators, the trading system avoids arbitrary guesses and operates on mathematical probability buffers. Extraneous noise has been removed to keep execution fast and deterministic.

- **Status:** Step 1 (Data Collection) and Step 2 (Signal & Indicator Engineering) are fully verified.
- **Next Document:** Step 3 (L2 Order Book Execution Simulator & Backtesting Validation).